

The Interplay of Race/Ethnicity and Parental Education in Educational Attainment: A Cohort Analysis

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Educational attainment is a key predictor of life outcomes. Despite this, access to opportunity within the American education system remains inequitable. This study examines how the intersectionality of race/ethnicity and parental education shape educational attainment among U.S. adults born between 1980 and 1984, using data from the National Longitudinal Survey of Youth 1997 (NLSY97). Prior research has highlighted racial disparities in school funding, disciplinary outcomes, and postsecondary access, as well as the effects of parental education on individual success. However, the interaction between the aforementioned variables remains under-explored in cohort-based analyses. This study applied multiple imputations and numerous logit models to investigate how educational attainment of residential parents, combined with race, influenced respondents' highest educational outcome completed by ages 33-37. Results indicate that while increased parental education is positively correlated with respondent educational attainment overall, the strength of this relationship varies significantly by race/ethnicity-parental interactions. Notably, Hispanic respondents demonstrated higher baseline odds of attainment but received smaller marginal benefits from increased parental education.

Keywords: Educational Attainment, Structural Inequality, Race, Parental Education

Introduction

School serves as one of the most important institutions within an individual's life, affecting development and socialization (Conley, 2019, p.551-555). Throughout this experience, students of color are subjected to ethnic-racial discrimination. According to the UN Human Rights Office of the High Commissioner (OHCHR) (1965), ethnic-racial discrimination refers to negative behaviors experienced by individuals who identify as people of color. While often conceptualized as individual behavior, these experiences are rooted in systemic structures that disproportionately allocate social capital and resources. The effects of this discrimination are varied in reach and impact,

from educational outcomes to adjustment within a school environment (Bottiani et al., 2017; Del Toro et al., 2024; Pearman, 2022).

Additionally, parental educational attainment has been shown to impact their children's ability to succeed within an educational setting (Assari, 2018; Oakes, 1990; Reed, 2005). This is made worse by the intersection of race (Assari, 2018). As such, it's important to consider how societal forces and structures affect differing educational outcomes, particularly with regard to parental educational achievement and ethnic-racial discrimination. Specifically, this analysis looks to answer the following question: among U.S. adults born from 1980 to 1984, how does the intersection of race/ethnicity and parental education af-

fect the highest educational milestone completed by ages 33-37? This question is of particular interest as educational inequality is rampant in the United States and can affect all facets of life, from wealth to health (Lurie et al., 2021; Zajacova & Lawrence, 2018). To answer this question, data from the National Longitudinal Survey of Youth 1997 (NLSY97), a comprehensive and nationally representative longitudinal survey, was analyzed utilizing multiple logit and binary regressions.

The Literature

Availability of Resources

The degree to which students and educators have access to resources significantly impacts student outcomes. Condrón and Roscigno (2003) found a positive correlation between school spending and test scores, with schools serving predominantly minority and low-income populations consistently underfunded. Looking to the National Center for Education Statistics (2023), over 30% of Hispanic, Black, and Native students attend low-income schools, compared to only 7% of White students. This stark disparity is often attributed to broader patterns of social stratification within U.S. society, with concrete examples within California's education system ((Kozol, 1991; Reed, 2005)).

Oakes (1990) argued that these inequities arise because inner-city schools lack the political influence needed to secure adequate funding. As a result, schools with a predominant minority student population often face severe resource deficits in science, thus limiting access to experienced teachers, adequate facilities, and technology necessary for academic success. It's notable that this article was prompted by a need for human capital—more workers in science and technology—and only partially driven by the intent to discover racial inequities.

Belonging Within Schools

Bottiani et al. (2017) completed a study that showed that gaps between White and Black stu-

dents within out-of-school suspension (OSS) had a significant negative correlation on sense of belonging within a Black student's school. Alongside this was a positive correlation in adjustment problems with higher Black suspension rates. A higher low-SES population within a school also contributed to these findings, highlighting the importance of resources in student success. King (2011) found the opposite: school discipline had no effect on belonging and outcomes based upon race. This study is limited in scope, however, as it only examines whether there was discipline. There was no analysis of whether significant gaps existed based upon race.

Georgiades et al. (2013) provided an interesting set of results in their study on belonging in school, emotional issues, and adjustment issues related to racial congruence and immigration generational status. For Asian and White students, an increase in congruence resulted in a negative correlation on behavioral problems, while the opposite was true for Black students (all groups being 3rd-generation immigrants). For emotional problems, we see a negative correlation for White and Black students when congruence increases, but a positive correlation for Asian students. The overall effects of congruence are very mixed. However, they are attenuated significantly by a sense of belonging, as this contributes directly to emotional and behavioral adjustment. That is to say, a sense of not belonging explains emotional and behavioral issues at a statistically significant level when a positive correlation is observed, rather than congruence.

Both studies contain parallels to critical race theory (CRT) and strain theory (Bottiani et al., 2017; Delgado & Stefancic, 2020; Georgiades et al., 2013). Looking upon CRT, systemic inequalities, whether it be unequal representation in the student body, school punishment, or economic resources, increase racially divided outcomes in school performance, perpetuating negative stereotypes and increasing inequalities outside of schooling environments. Within strain theory, the pressures of belonging and increased punishment create strain

upon students' psyches and performance, resulting in the findings of Georgiades et al. (2013). Additionally, taking into account the social capital of parental education, lower parental attainment serves to increase the negative effects of both strain and critical race theory.

School Achievement

Testing in Primary and Secondary Settings

The Black-White score gap is a numerical manifestation of systemic inequities within education and society. Pearman (2022) found that Black students are performing upwards of 0.5 standard deviations (SD) lower than their White peers, with this increasing modestly (0.1-0.2 SDs) with a SD of 1 or more with regard to county-level implicit and explicit bias. Del Toro et al. (2024) found similarly that racist encounters with an educator, specifically in math, lead to lower test scores in both victimized students and students aware of the aggression. Students experiencing vicarious discrimination were subject to even lower scores than students directly affected, though the converse is true for class grades.

The score gap is also directly related to resources, where every additional \$1,000 in funding per student accounts for a subject-dependent increase of 6%-10% on state-administered standardized tests (Condrón & Roscigno, 2003; Reed, 2005). Reed (2005) found that clearance rates for the 10th grade exit test in California were appreciably lower in every race except Asians when compared to White students. Furthermore, over half of all Hispanic students were in low-performing schools, where there were 7% fewer properly credentialed educators on staff.

Post-Secondary Achievement and Family

Reed (2005) found data supporting the concept that generational achievement affects outcomes for minority students. There is a correlation between the completion of secondary and post-secondary education by a mother and their four-year-old's

likelihood to attend early childhood education (ECE) programs. According to Jones (2024), there are significant benefits associated with attendance of ECE programs, amounting to higher wages and higher completion of education.

Moving back towards Reed (2005), she finds that attenuating factors include general SES level, though SES is predicated often upon academic achievement. The general trend shows a vicious cycle of lower educational outcomes begetting lower educational outcomes. Assari (2018) undertook a very similar study to the current one, finding that between Black and non-Hispanic White individuals, educational attainment was greater for White individuals, but educational mobility was higher for Black individuals. Furthermore, Assari included covariates for gender, finding these effects only occurred for female respondents.

Theoretical Framework

Racial/Ethnic Generational Achievement & Inequalities

Through the late 1980s and early 1990s, Kimberlé Crenshaw (1989; 1991) developed a framework for analyzing the intersecting and additive effects of inequalities called Intersectionality. Intersectionality posits that simply examining inequalities experienced by an individual or group one-by-one is inadequate; rather, looking at how they work in tandem better explains the experiences and outcomes observed.

Additionally, a preeminent theory that attempts to explain segmentation in immigrant class outcomes and mobility is segmented assimilation theory, proposed by Portes and Zhou (1993). In their paper, Portes and Zhou examined second-generation immigrant households, arguing that not all immigrant groups follow the same upward trajectory of class mobility. Instead, assimilation pathways are shaped by numerous structural factors, such as race and community support structures. This often leads to outcomes that diverge in effects, from downward class mobility to as-

similation of cultural ideas. These theories combined provide a valuable framework from which to observe the impact and effects of generational achievement. In particular, they help to explain how generational achievement may impact educational attainment differently depending on the social and institutional context in which families and individuals are embedded.

Hypothesis

From the available research and theory, the expected outcomes of the current study are as follows:

- H₁: Higher levels of parental education and identifying as non-Black/non-Hispanic will independently lead to higher educational milestones completed by ages 33–37.
- H₂: The positive association between parental educational attainment and a respondent's own educational outcome will be significantly stronger for White individuals than for individuals of color.
- H₃: Specifically, it is expected that at higher levels of parental education, the attainment gap between non-Black/non-Hispanic respondents and Black or Hispanic respondents will persist or widen, suggesting that systemic racial discrimination attenuates the social capital typically provided by high parental education.

Methods

NLSY97 Survey Design, Sampling, and Participants

As previously described, the sample used was the NLSY97 conducted by the US Bureau of Labor Statistics (BLS) (2024). This is a nationally representative sample carried out every year pre-2011 and every two years thereafter, with selection finalized in 1997 and round 1 of questions carried out in 1998. This database was chosen due to the large

sample size ($n = 8,984$), robust and comprehensive question selection, representative nature (being weighted and stratified), and ease of access.

Individuals were chosen who were born between 1980 and 1984. At the time of first interviews, respondents' were ages 12 to 16. The sampling strategy, as described by the BLS, is as follows:

“The NLSY97 cohort was selected in two phases... In the first phase, a list of housing units for the cross-sectional sample and the over-sample was derived from two independently selected, stratified multistage area probability samples. This ensured an accurate representation of different sections of the population defined by race, income, region, and other factors. In the second phase, sub-samples of the eligible persons identified in the first phase were selected for interview.”

The longitudinal sample was used for all data analysis. Further information about the NLSY97 is available on the NLS website (2024).

Measures

Respondent Education

Within the NLSY97 (2024), the data was subset to specify participants in round 18 who responded with an answer to the following question: “[What is] the highest degree received as of the survey date?” This was on a scale of 0 to 7, with zero representing no high school completion and 7 representing a professional degree. All responses falling outside this range (non-interviews and invalid skips for respondents, ungraded for parents) were removed. This accounted for a removal of $n = 2283$, with most being non-longitudinal respondents. Important to note is the difference in scales between parental and respondent education levels (see appendix A, table A1 for the complete codebook).

Looking further into respondents HGC, there were two pairs of responses that posed a unique question with how to code: High School Diploma-GED and PhD-Professional degree (PD). Each pair provides similar results; a High School Diploma (HSD) and a GED both convey completion of grades K-12, whilst a PhD and PD are both terminal degrees. However, a HS Diploma and GED affect an individual's life course in differing ways, largely due to social stigma and age differences in attainment when compared to a diploma (Maralani, 2011). As such, HSD-GED were kept separated. On the contrary, the trajectory of PhD and PD are similar and were thus collapsed into a single response value. This resulted in a 0-6 scale. Additionally, any analysis that required removing the ordered factor structure from the variable resulted in a scale shifted by +1.

Parental Education

For parental education, residential parent (separated into mother and father - coded as "mom" and "dad") was chosen over biological, as the present study is concerned about the social power of education, not possible biological factors. As for coding, multiple approaches were considered. An analysis with parental education kept separate (additive effects) was tested but yielded results with high collinearity, suppression effects, and extremely high Variance Inflation Factor (VIF). For this reason, parental education was coded as the maximum between each residential pair. Where a residential parent was missing, education was coded as the education of the present parent. If both were missing, the response was coded as missing (NA).

Race and Ethnicity

The NLSY97 offers a number of variables encoding race and/or ethnicity. For this analysis, the self-identified race/ethnicity key variable from round 1 was utilized. Round 4 contains seven dummy variables encoding race for the longitudinal sample; however, this increases the number

of covariates sizeably and is interviewer-identified, whereas round 1 is self-identified. By forgoing the data from round 4, overfitting and implicit interviewer bias are avoided.

To test for effects, three dummy variables were coded (figure B) for race/ethnicity, with non-Black/non-Hispanic (NBNH) as the reference variable. This was done prior to imputation and was included in the predictor set.

Imputations

A unique aspect of the data that made analysis more complex was the number of answers missing from either one or both of the residential parent's educational attainment variable (round 1) (see appendix A). Figure B4 presents the missing data using an upset plot. An earlier analysis was completed with separated parent HGC values, which required imputation due to a casewise missingness of $n = 2686$. After combining mother and father HGC to a single variable, missingness was reduced to 9% or $n = 79$.

Despite this reduction, imputations have been shown to produce better analyses than case-wise deletion, especially when the data is MAR (Afghari et al., 2019). As such, Jamshidian and Jalal's (2010) non-parametric Missing Completely at Random (MCAR) test was used to determine whether the pattern was random. Additionally, a number of sensitivity analyses were ran, ranging from regressing missing flags on observed values to comparing mean and SD values across parental pairs when accounting for missingness (See appendix for these analyses). These tests presented sufficient evidence for Missing At Random (MAR). Thus, all analyses proceeded under a MAR assumption.

Following this, data was imputed using the *MICE* package, which utilizes Gibbs sampling (Buuren, 2021). The imputation method chosen was predictive mean matching (PMM), as PMM provides plausible data that is robust given multiple data types (Buuren, 2021). Twenty imputations ($m = 20$) were conducted, with the model matrix

including every covariate (see equation 1) and the dependent variable.

Models

All models featured the NLSY97 complex survey design, using sampling weights, strata, and primary sampling units (PSUs) via the *svy* and *svyVGAM* frameworks. Models were estimated and then pooled across imputations using Rubin's rules (Buuren, 2021, Section 5.2).

The primary model was a survey-weighted Proportional-odds Ordinal Logistic Regression (POLR). The following regression equation was used:

$$\begin{aligned} \text{logit}[P(Y_i \leq k)] = & \theta_k - \beta_1 \text{HGCParent}_i \\ & + \beta_2 \text{Black}_i + \beta_3 \text{Hispanic}_i + \beta_4 \text{Mixed}_i \\ & + \beta_5 (\text{HGCParent}_i \times \text{Black}_i) \\ & + \beta_6 (\text{HGCParent}_i \times \text{Hispanic}_i) \\ & + \beta_7 (\text{HGCParent}_i \times \\ & \text{Mixed}_i), \quad k = 1, \dots, K - 1. \quad (1) \end{aligned}$$

A secondary model featuring a quasibinomial logit was also tested for robustness checks. A passing value (1) was coded as achieving a degree of BA or higher. Quasibinomial was chosen over a standard binomial logit regression as the implementation of the standard binomial family in R fails to handle weights correctly. The following regression equation was used:

$$\begin{aligned} \text{logit}[P(\text{binaryHGC}_i = 1)] = & \alpha + \\ & \beta_1 \text{HGCParent}_i + \beta_2 \text{Black}_i + \beta_3 \text{Hispanic}_i \\ & + \beta_4 \text{Mixed}_i + \beta_5 (\text{HGCParent}_i \times \text{Black}_i) \\ & + \beta_6 (\text{HGCParent}_i \times \text{Hispanic}_i) \\ & + \beta_7 (\text{HGCParent}_i \times \text{Mixed}_i). \quad (2) \end{aligned}$$

Notably, stratum 117 presented only one PSU as a side effect of subsetting the data for model training. An analysis with the full dataset was conducted and presented similar results; as such, the training set was utilized and the mean for this stratum was set to the population mean.

Table 1

Number of Respondents and Highest Grade Completed by Race.

HGC	Race	<i>n</i>	Mean	SD
None	Black	174	127.75	84.06
	Hispanic	161		
	Mixed	2		
	NBNH ¹	174		
GED	Black	328	215.25	154.99
	Hispanic	179		
	Mixed	11		
	NBNH	343		
HS	Black	796	672.5	495.84
	Hispanic	655		
	Mixed	21		
	NBNH	1218		
AA	Black	144	149.5	131.23
	Hispanic	121		
	Mixed	8		
	NBNH	325		
BA	Black	256	338	369.6
	Hispanic	210		
	Mixed	16		
	NBNH	870		
MA	Black	99	135	164.71
	Hispanic	62		
	Mixed	4		
	NBNH	375		
PhD*	Black	30	37.25	45.31
	Hispanic	14		
	Mixed	2		
	NBNH	103		

Note. *PhD and PD. ¹NBNH = Non-Black Non-Hispanic.

Results

Descriptive Statistics

In total, 6701 individuals were included in the analysis from a possible 8,984. Of this, 50.9%

Table 2

Missing Data Patterns for Jamshidian and Jalal's MCAR Test.

Pattern	Value is Missing?			<i>n</i>
	HGC Res	HGC Mom	HGC Dad	
1	Y	N	N	79
2	Y	Y	Y	588
3	Y	Y	N	296
4	Y	N	Y	36

Note. HGC = Highest Grade Completed.

were White ($n = 3408$) with a mean education of 3.85, 27.3% were Black ($n = 1827$) with a mean education of 3.22, 20.9% were Hispanic ($n = 1402$) with a mean education of 3.20, and 1.0% were mixed ($n = 64$) with a mean education of 3.70. The mode of education across the board was 3 (HS Diploma). Table 1 provides a breakdown, by race, of educational attainment of all respondents, alongside educational category averages.

Inferential Statistics

When the data for parental education is separated, Jamshidian and Jalal's test detects the data is not MCAR, and significantly so (Table 2). This provides partial evidence to suggest the data is MAR and has a pattern predicted by an observed variable(s). Secondly, mean and SD was relatively stable across categories (Table 3). Additionally, dad and mom missingness was significantly correlated with race. Dad missingness was also correlated with mom HGC; however, reversing the regression lead to no correlation. Besides the last regression, these results suggest a MAR mechanism. Additionally, assuming MNAR, when HGC was pooled for parents, it lead to a relatively small n of cases where this MNAR mechanism could occur. Finally, sensitivity analyses on the POLR model revealed model stability across assumptions. Thus, the decision to continue using imputations was made.

Table 3

Sensitivity Analysis of Means and SD Against Father Missingness

	Dad HGC		Mom HGC		Race	
	Mean	SD	Mean	SD	Mean	SD
Missing	NA	NA	12.1	2.7	2.2	1.3
Observed	12.9	3.3	12.8	3.0	3.1	1.3
Mom Miss						
Missing	0.2	0.4				
Observed	0.1	0.2				

Note. HGC = Highest Grade Completed.

It is important to note that any time an imputation is used, there is a chance bias and variability will be introduced which may not have existed in the dataset prior to the imputation (Buuren, 2018). However, the benefits of an imputation in the present study—capturing data, especially in categories with low response numbers—outweigh the drawbacks. The relative homogeneity in density between the imputed and observed values (Figures B5) supports the quality of the imputations, likely leading to less effect on the study outcome when compared to casewise deletion.

POLR Model

The results of the pooled POLR model (figure 4) reveals that three of the seven predictors are statistically significant at $CI = 95\%$, those being HGC of the residential parent ($p < 0.01$, $\alpha = 0.05$), the interaction effect of residential parent HGC and being Hispanic ($p < 0.01$, $\alpha = 0.05$), and being Hispanic ($p < 0.01$, $\alpha = 0.05$) (table 2). The effects of residential HGC is in line with the literature, with an odd ratio slightly above 1. This means the effect is rather mild, but still noticeable and prevalent.

Notably, being Hispanic is indicative of increased log odds. Furthermore, Hispanic residential parents see a decreased effect of parental education increasingly respondents educational attainment, with an odd ratio of 0.84 ($p < 0.01$,

Table 4*Survey-Weighted Proportional-Odds Logit Regression*

	Log Odds	Odd Ratio	SE	Statistic ¹	95% CI	
					2.5 %	97.5 %
HGCParent	0.324*	1.382	0.015	21.719	0.294	0.353
Black	-0.274	0.760	0.533	-0.514	-1.335	0.788
Hispanic	2.032*	7.633	0.299	6.792	1.442	2.623
Mixed	-0.850	0.427	1.421	-0.598	-3.661	1.960
HGCParent:Black	-0.013	0.987	0.041	-0.311	-0.094	0.069
HGCParent:Hispanic	-0.167*	0.846	0.023	-7.330	-0.212	-0.122
HGCParent:Mixed	0.021	1.022	0.089	0.240	-0.155	0.198
None GED	1.356*	-	0.218	6.223	0.926	1.786
GED HS	2.617*	-	0.209	12.523	2.205	3.029
HS AA	4.635*	-	0.217	21.316	4.206	5.064
AA BA	5.098*	-	0.220	23.127	4.663	5.532
BA MA	6.657*	-	0.238	27.986	6.187	7.126
MA PhD	8.361*	-	0.259	32.287	7.850	8.872

Note. HGC = Highest Grade Completed. ¹Wald Test. * $p < 0.05$

$\alpha = 0.05$).

Looking at the non-significant predictors in the POLR model, most have an intercept within ± 0.05 of 0, suggesting that, even if these predictors (all race-parent variables with the exception of Hispanic) were significant, their effects compared to the reference group, non-White/non-Hispanic individuals, are minimal. Black and mixed race show predictor estimates that have slightly stronger negative correlation, but each have extremely high p -values and confidence intervals overlapping 0.

Unsurprisingly, the threshold estimates for education show that almost every jump between educational levels have non-overlapping cutpoints, with the exception of HS|AA and AA|BA. This is possibly due to the strong emphasis on college completion in the US. Additionally, this lends credence to keeping GED and HS separate.

Quasibinomial Logit Model

The quasibinomial model (figure 5) presents slightly divergent results. Significance is similar across the board, with the exception of the interac-

tion of parental education and Mixed race, with a slightly negative log odd and marginally lower p -value ($p < 0.10$, $\alpha = 0.05$). Additionally, the predictive power of being Hispanic was lower when compared with the POLR model.

Discussion

Overall, the data partially supports and refutes the test hypotheses. Thus, H_1 is partially refuted, while H_2 and H_3 received mixed support depending on racial/ethnic group. While it's true that higher grade completion from a residential parent positively affects educational attainment, when interacting with race, White individuals do not see significant benefits of HGC from their parent over individuals of other races and ethnicities—with the exception of Hispanic individuals, who experience reduced benefits from parental HGC. However, when not accounting for parental HGC, Hispanic individuals have increased log-odds of achieving higher education.

Table 5*Survey-Weighted Quasibinomial Logit Regression*

	Log Odds	OR	SE	Statistic ¹	95% CI	
					2.5 %	97.5 %
(Intercept)	-5.697*	0.003	0.336	-16.962	-6.363	-5.031
HGCParentEd	0.365*	1.441	0.022	16.711	0.322	0.409
Black	0.443	1.557	0.716	0.619	-0.984	1.870
Hispanic	1.661*	5.263	0.538	3.087	0.593	2.728
Mixed	2.817	16.721	1.904	1.479	-0.986	6.620
HGCParent:Black	-0.059	0.942	0.052	-1.134	-0.164	0.045
HGCParent:Hispanic	-0.141*	0.868	0.040	-3.556	-0.220	-0.062
HGCParent:Mixed	-0.217†	0.805	0.120	-1.804	-0.457	0.023

Note. HGC = Highest Grade Completed. ¹Wald Test. * $p < 0.05$ † $p < 0.10$

Parental Education: A Universal Driver of Higher Education

Consistent with the literature, higher residential parental HGC significantly predicts increased educational attainment ($OR = 1.382, p < 0.01, \alpha = 0.95$) (Assari, 2018; Jones, 2024; Reed, 2005). While modest, this effect predicts that, with each additional grade completed by a parent, the odds that their child will progress to a higher educational milestone increases by 38%.

The literature on educational transmission suggests this effect should be stronger among White families (Assari, 2018; Oakes, 1990), yet the POLR and quasibinomial models show no significant effect differences by race for the Black and Mixed race groups. This pattern itself is notable: it suggests that while parental education matters universally, it matters in similar ways across racial groups.

Education and Identifying as Hispanic

Using segmented assimilation theory (Portes & Zhou, 1993), it is possible that the results we see from Hispanic individuals come from a multitude of societal and familial structures. Previous research shows Hispanic educational attainment (AA or above) to be severely below that of

many other racial groups in the US, particularly White (Kim et al., 2024). We are beginning to see increases in the rates at which Hispanic individuals attain higher education, and while these rates may not be high compared to the general population (48.1% versus 29.5%), segmented outcomes in the individuals studied can possibly explain the results. These segmentations could arise in multiple fashions. For example, this could be due to the citizenship status of individuals surveyed and whether this proportion is generalizable to the US population of Hispanic individuals. This may also align with findings on immigrant optimism, where high educational aspirations persist notwithstanding socioeconomic disadvantages (Kao & Tienda, 2022), potentially driving attainment independently of parental HGC.

Furthermore, Kao and Tienda (2022) found that the immigration status of Hispanic individuals is less important to educational achievement. This may be due in part from societal or cultural barriers which prevent the full relation of educational capital from parent to child in Hispanic families. The findings suggest that simply improving parental education levels may not equitably improve intergenerational mobility for all racial groups. Educational policy must then address these societal/cultural barriers and structures, should they be

present.

Interactions for Black and Mixed-Race Respondents

Contrary to H_2 and H_3 , Black respondents show no significant interactions between parental education and educational attainment ($OR = 0.987$, $p > 0.10$), and Mixed race respondents similarly show no significant effect in the POLR model. A possible explanation for these results lies in Bot-tiani et al. (2017), who found that discipline gaps negatively correlated with school belonging, mediated by school-wide SES. It's possible that the barriers depicted here are so pervasive they prevent the upwards effects of parental education from appearing.

The extremely small sample of Mixed race respondents ($n = 64$) makes interpretability difficult for this group, as noted in limitations. However, the pattern of non-significance across both Black and Mixed groups suggests that intersectionality operates selectively between the reference group and focus groups; it emerges strongly for Hispanic respondents but not for other groups. This mechanism warrants further research into the conditions under which intersectional processes are observable or significant in quantitative data.

Limitations

As previously noted, imputations can introduce error into analyses from inconsistent, variable data, possibly affecting results. Furthermore, measures drawn from imputed data rely upon proper model specification and a MAR or MCAR assumption. However, given the percent of missing and density graph, there is sufficient evidence to support the quality of the imputation. Additionally, an increased amount of runs serves to reduce variability in the observed effects of fitted models.

Furthermore, GED and HS are interpreted as discrete ordinal categories, with GED placed below HS. While societally they may be seen as different, practically, they hold similar functional dis-

inction. On the other hand, a PhD and a professional degree were collapsed into a single level (level 7). See *methods*.

Looking at the various models, a lack of additional covariates serves to prevent overfitting; however, it can also prevent the full picture from becoming evident. Given the relatively high VIF present in the current study, it's recommended that future papers utilize dimensionality reduction techniques or regularization with an increase in covariates.

Lastly, the sample size for individuals of mixed race is excessively small ($n = 64$) after selecting valid cases and prior to casewise deletion or imputation of invalid/missing parent HGC answers. As such, large uncertainty and low prediction validity was expected. Future studies should aim to encapsulate a greater number of respondents within this demographic, as it is underexplored.

Implications for Future Research

Future research should expand upon the present study by including confounders which may create more generalizable results, such as increasing covariates. Additionally, a longitudinal study tailored specifically to researching the intersectionality of race and parental education may yield better results. Segmenting race and ethnicity further may allow a more nuanced view of racial-ethnic divides in attainment. Furthermore, future research may seek to define educational success along the lines of educational achievement and job status, as education is only one small part in a person's wellbeing. With regards to ethnic and racial groups, future studies should attempt to identify societal and cultural factors which moderate gains or decreases in attainment.

Conclusions

This study contributes to the educational inequality literature by presenting that not only do disparities exist, but how and for whom educational capital strengthens outcomes across gen-

erations. The results demonstrate that intersectionality operates selectively, with results and significance that vary by racial/ethnic group. Hispanic respondents present an intriguing outcome of high baseline attainment coupled with restrained parental education effects, possibly reflecting immigrant optimism and within-group heterogeneity. Black respondents show uniform constraint across all parental education levels, consistent with theories of pervasive structural racism. These patterns highlight that reducing educational inequality requires moving beyond individual- or family-level interventions to address the institutional structures that shape opportunity inequitably across racial and ethnic groups.

References

- Afghari, A. P., Washington, S., Prato, C., & Haque, M. M. (2019). Contrasting case-wise deletion with multiple imputation and latent variable approaches to dealing with missing observations in count regression models. *Analytic Methods in Accident Research*, 24, 100104. <https://doi.org/10.1016/j.amar.2019.100104>
- Assari, S. (2018). Parental Education Attainment and Educational Upward Mobility; Role of Race and Gender [Number: 11]. *Behavioral Sciences*, 8(11), 107. <https://doi.org/10.3390/bs8110107>
- Bottiani, J. H., Bradshaw, C. P., & Mendelson, T. (2017). A multilevel examination of racial disparities in high school discipline: Black and white adolescents' perceived equity, school belonging, and adjustment problems. *Journal of Educational Psychology*, 109(4), 532–545. <https://doi.org/10.1037/edu0000155>
- Buuren, S. v. (2021). *Flexible imputation of missing data* (Second edition). Chapman & Hall/CRC.
- Condon, D. J., & Roscigno, V. J. (2003). Disparities within: Unequal Spending and Achievement in an Urban School District. *Sociology of Education*, 76(1), 18. <https://doi.org/10.2307/3090259>
- Conley, D. (2019). *You may ask yourself: An introduction to thinking like a sociologist* (Sixth edition). W.W. Norton.
- Crenshaw, K. (1989). Demarginalizing the Intersection of Race and Sex: A Black Feminist Critique of Antidiscrimination Doctrine, Feminist Theory and Antiracist Politics. 1989(1).
- Crenshaw, K. (1991). Mapping the Margins: Intersectionality, Identity Politics, and Violence against Women of Color. *Stanford Law Review*, 43(6), 1241–1299. <https://doi.org/10.2307/1229039>
- Del Toro, J., Legette, K., Christophe, N. K., Pasco, M., Miller-Cotto, D., & Wang, M.-T. (2024). When ethnic-racial discrimination from math teachers spills over and predicts the math adjustment of nondiscriminated adolescents: The mediating role of math classroom climate perceptions. *Developmental Psychology*, 60(12), 2242–2257. <https://doi.org/10.1037/dev0001833>
- Delgado, R., & Stefancic, J. (2020, December). *Critical Race Theory (Third Edition): An Introduction*. New York University Press. <https://doi.org/10.18574/nyu/9781479851393.001.0001>
- Georgiades, K., Boyle, M. H., & Fife, K. A. (2013). Emotional and Behavioral Problems Among Adolescent Students: The Role of Immigrant, Racial/Ethnic Congruence and Belongingness in Schools. *Journal of Youth and Adolescence*, 42(9), 1473–1492. <https://doi.org/10.1007/s10964-012-9868-2>
- Jamshidian, M., & Jalal, S. (2010). Tests of Homoscedasticity, Normality, and Missing Completely at Random for Incomplete Multivariate Data. *Psychometrika*, 75(4), 649–674. <https://doi.org/10.1007/s11336-010-9175-3>

- Jones, C. (2024). Benefits of Early Childhood Education on Later Adult Life. *Capstone Projects and Master's Theses*. https://digitalcommons.csumb.edu/caps_thes_all/1721
- Kao, G., & Tienda, M. (2022, January 12). Optimism and Achievement: The Educational Performance of Immigrant Youth*. In *Interdisciplinary Perspectives on the New Immigration* (1st ed., pp. 83–101). Routledge. <https://doi.org/10.4324/9781315054216-4>
- Kim, J. H., Claudia, M., Zhao, Z., & Swirsky, E. (2024). Race and Ethnicity in Higher Education: 2024 Status Report. Retrieved February 14, 2026, from <https://www.equityinhighered.org/resources/report-downloads/race-and-ethnicity-in-higher-education-2024-status-report/>
- King, L. (2011). An Examination of the Influence of School Discipline, Race/Ethnicity, and Gender on Students' Attitudes Toward School. *Theses and Dissertations*. <https://scholarcommons.sc.edu/etd/828>
- Kozol, J. (1991). *Savage Inequalities* [OCLC: 24821565]. Brilliance Corp.
- Lurie, L. A., Hagen, M. P., McLaughlin, K. A., Sheridan, M. A., Meltzoff, A. N., & Rosen, M. L. (2021). Mechanisms linking socioeconomic status and academic achievement in early childhood: Cognitive stimulation and language. *Cognitive Development*, 58, 101045. <https://doi.org/10.1016/j.cogdev.2021.101045>
- Maralani, V. (2011). From GED to College: Age Trajectories of Nontraditional Educational Paths. *American educational research journal*, 48(5), 1058–1090. <https://doi.org/10.3102/0002831211405836>
- National Center for Education Statistics. (2023). Concentration of Public School Students Eligible for Free or Reduced-Price Lunch. Retrieved February 14, 2026, from <https://nces.ed.gov/programs/coe/indicator/clb>
- Oakes, J. (1990). *Multiplying inequalities: The effects of race, social class, and tracking on opportunities to learn mathematics and science*. Rand.
- of Labor Statistics, B. (n.d.). NLSY97 Data Overview. Retrieved February 14, 2026, from <https://www.bls.gov/nls/nlsy97.htm>
- Pearman, F. A. (2022). Collective Racial Bias and the Black-White Test Score Gap. *Race and Social Problems*, 14(4), 283–292. <https://doi.org/10.1007/s12552-021-09347-y>
- Portes, A., & Zhou, M. (1993). The New Second Generation: Segmented Assimilation and its Variants. *The ANNALS of the American Academy of Political and Social Science*, 530(1), 74–96. <https://doi.org/10.1177/0002716293530001006>
- Reed, D. (2005). Educational Resources and Outcomes in California, by Race and Ethnicity.
- Zajacova, A., & Lawrence, E. M. (2018). The Relationship Between Education and Health: Reducing Disparities Through a Contextual Approach. *Annual Review of Public Health*, 39(1), 273–289. <https://doi.org/10.1146/annurev-publhealth-031816-044628>

Appendix A Codebook

Table A1

Codebook of Variables.

Var	Key	Var	Key
HGC Mom & Dad		*Race & Ethnicity	
	0		1
	1		2
	2		3
	3		4
	4		
	5		
	6		
	7		
	8		
	9		
	10		
	11		
	12		
	13		
	14		
	15		
	16		
	17		
	18		
	19		
	20		

HGC = Highest Grade Completed. NBNH = Non-Black Non-Hispanic. *These variables are raw NLSY97 data and are mutated prior to use in the paper.

Appendix B Expanded Demographics and Figures

Appendix C Missing Mechanism Analyses

Figure B1

Highest Grade Completion of Respondant by Race.

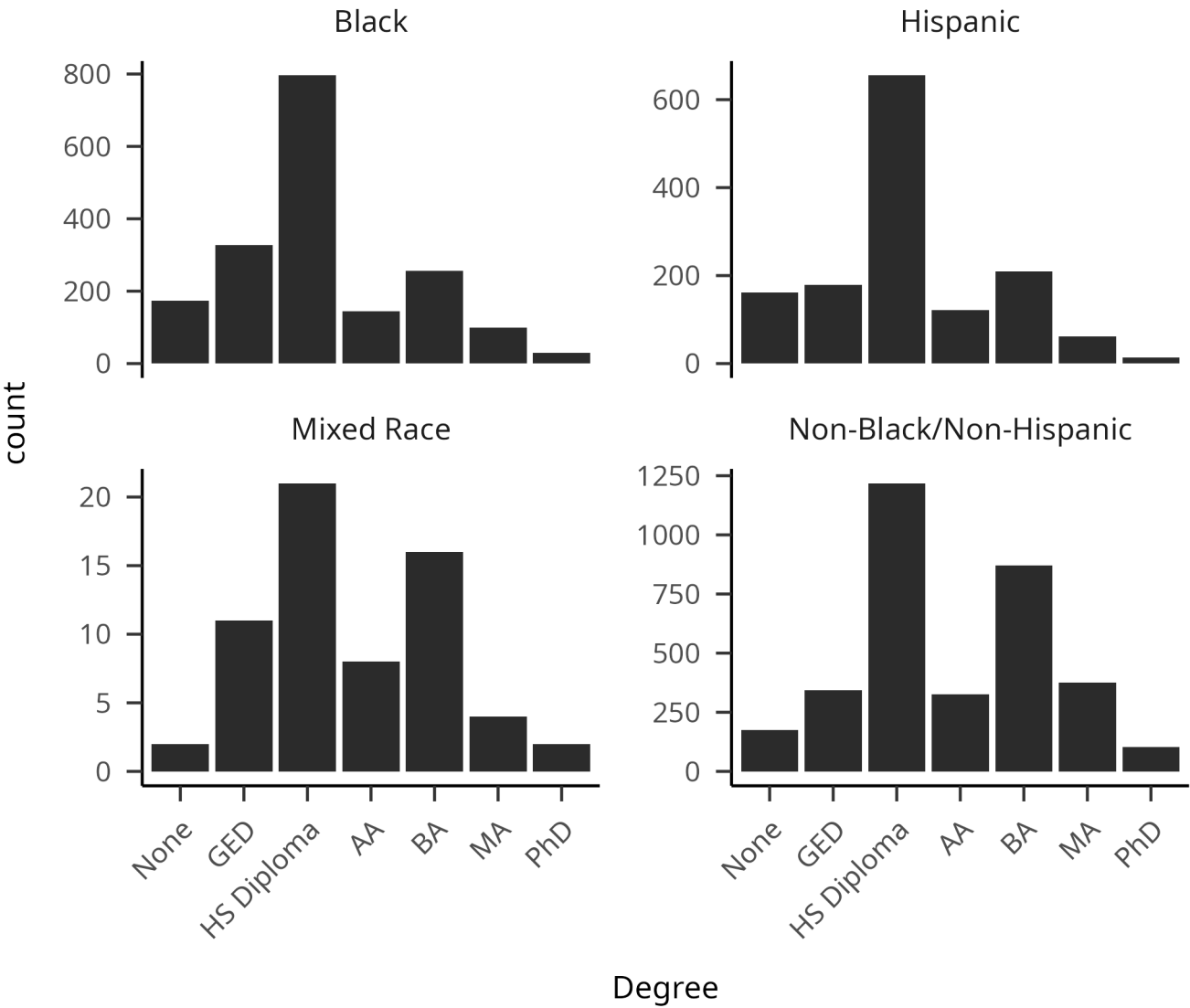


Figure B2

Highest Grade Completion of Residential Mom by Race.

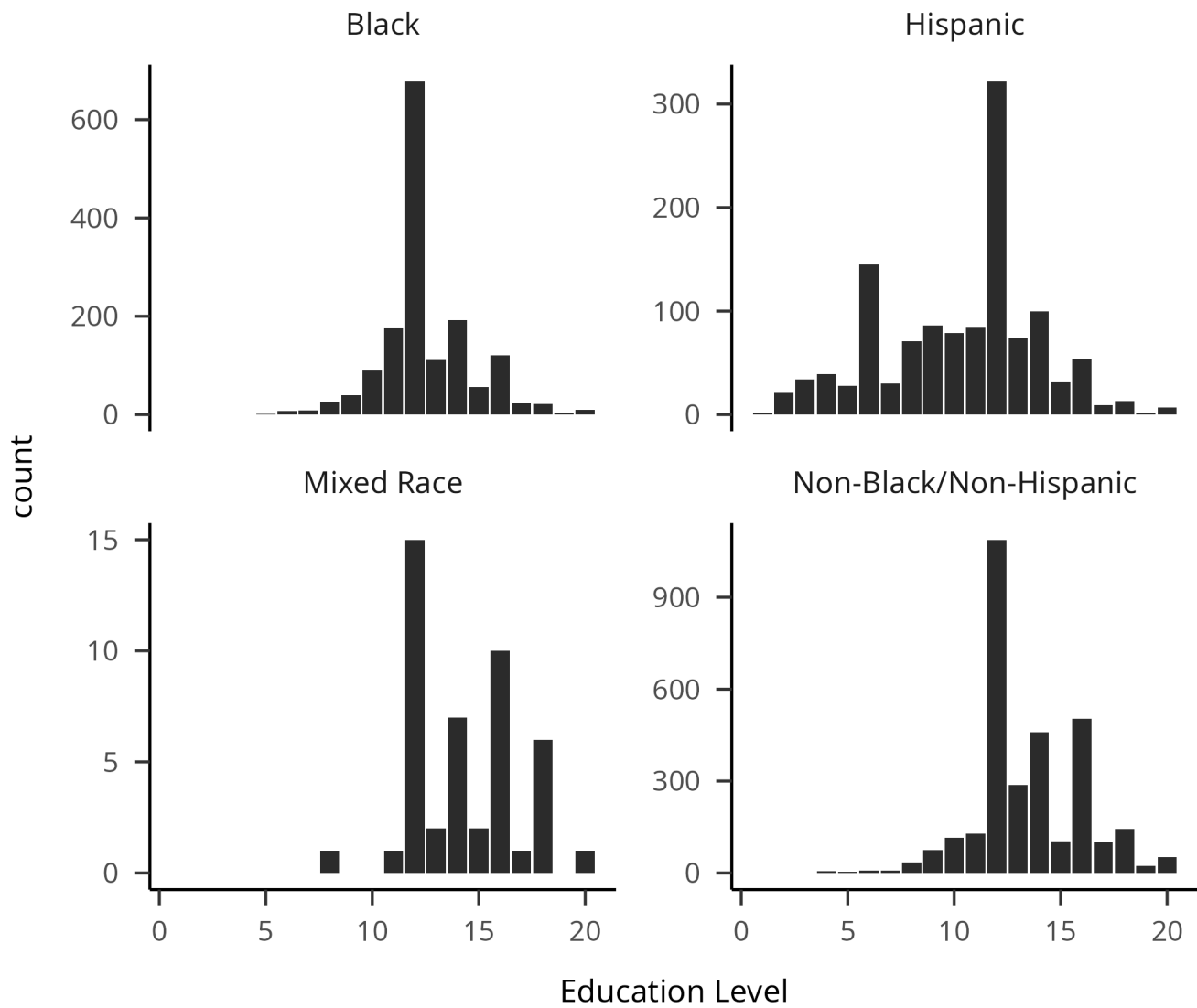


Figure B3
Highest Grade Completion of Residential Dad by Race.

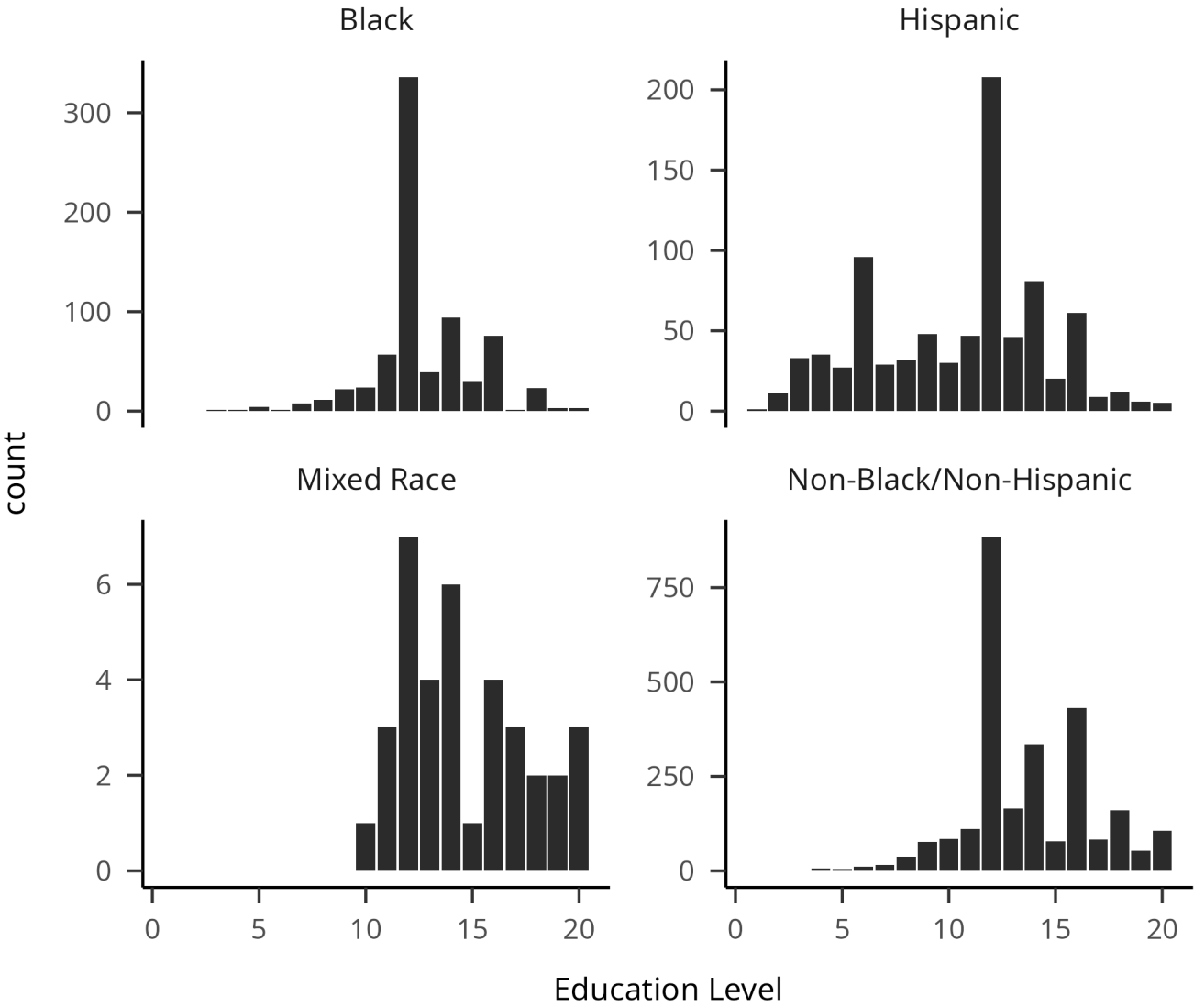
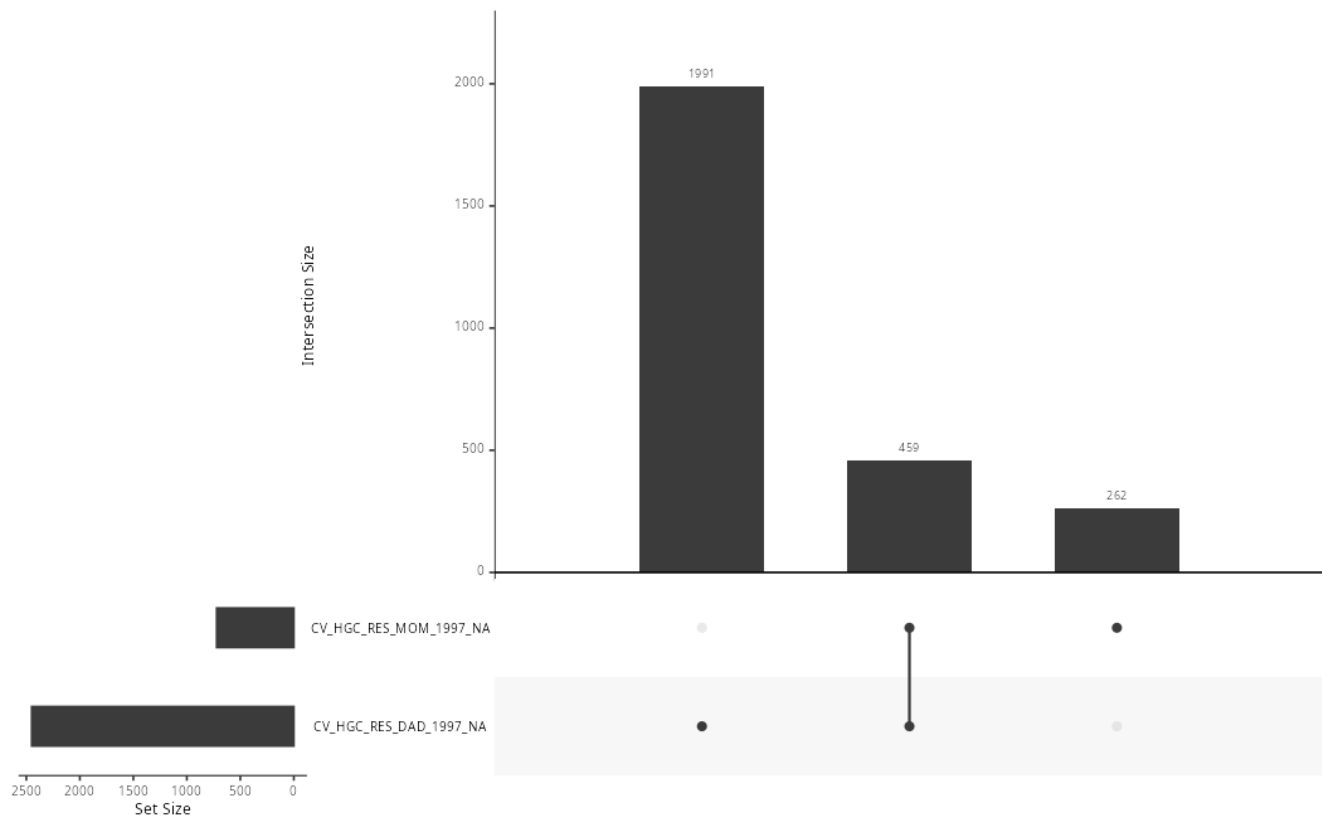


Figure B4

Upset plot of Missing Parental Education Values.

**Figure B5**

Imputation Vs Observed Value Density.

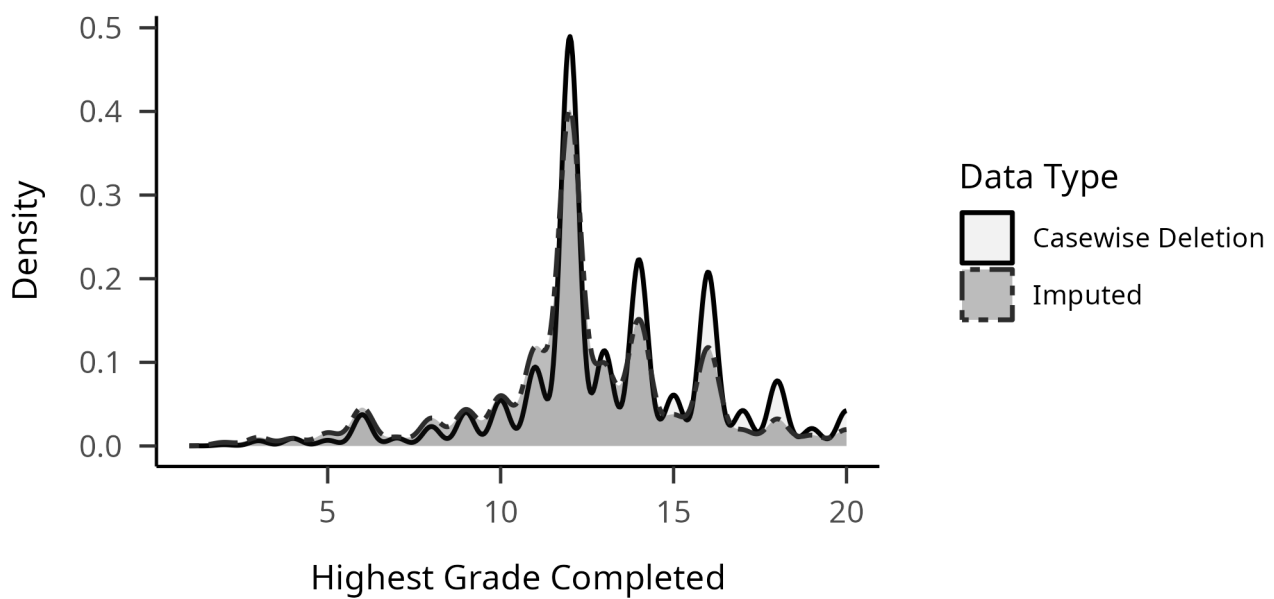


Figure B6

Log Odd Estimates of Educational Attainment with 95% Confidence Intervals.

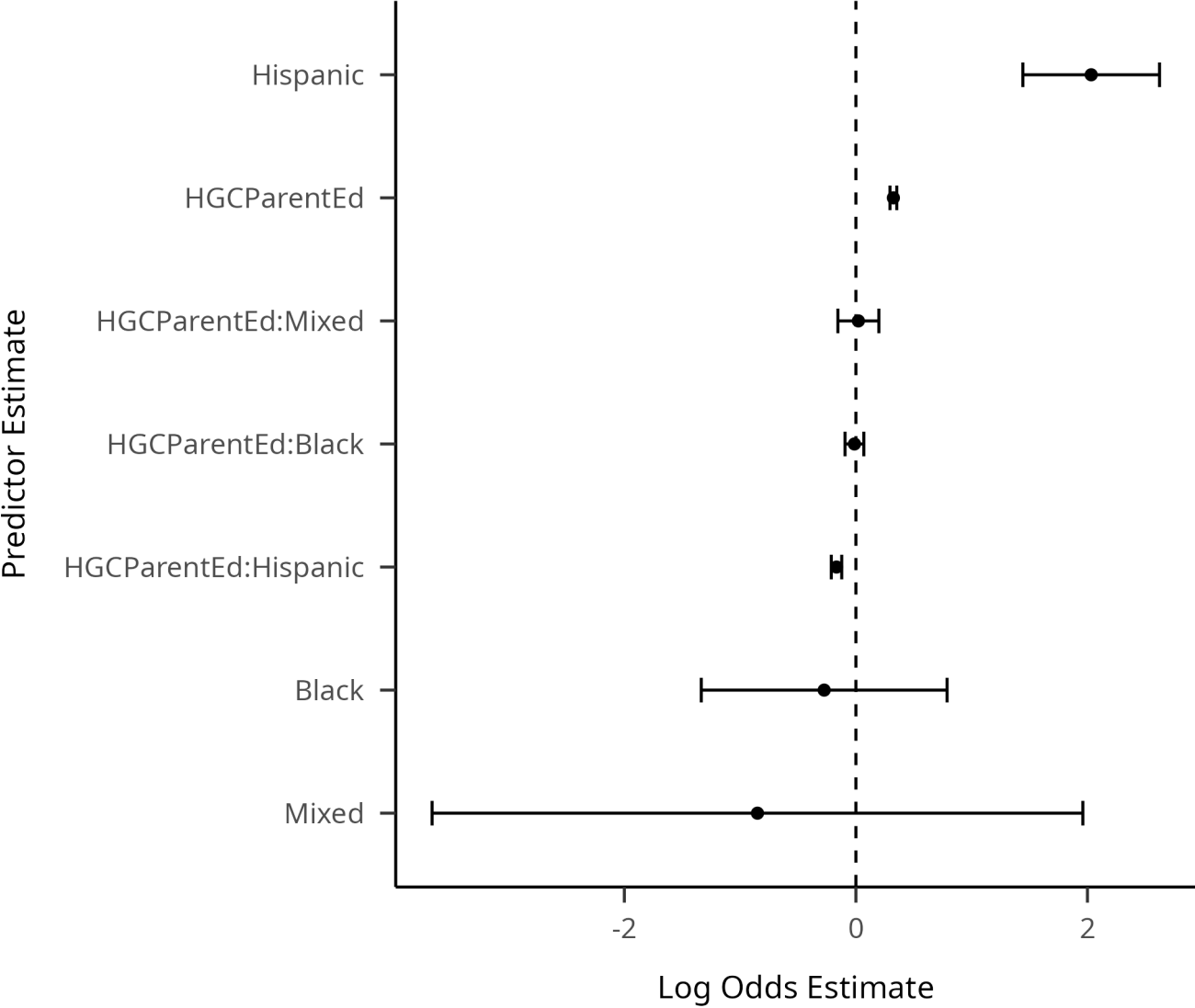


Figure B7

Cutpoint Estimates of Educational Attainment with 95% Confidence Intervals.

